

DATUM

Intent Pipeline

From raw browsing behavior to a predictive intent score, per session and per customer.

Project overview • Data layer, analytics rollups, and the V1 intent scoring engine

Prepared as an internal technical overview.

1. What This System Does

The Intent Pipeline turns raw, anonymous browsing activity on the storefront into three increasingly useful things: a clean historical record of what happened, daily analytics that describe engagement patterns, and a predictive score that estimates how likely a specific visit — or a specific returning customer — is to convert.

It is built as three layers that hand off to each other in order. Each layer reads only from what the previous layer already finished writing, so a failure in one layer never corrupts the others, and every layer can be re-run safely without producing duplicate or incorrect results.



In one sentence

Every browsing session gets scored 0–100 on engagement, classified as Low / Medium / High relative to that day's own traffic, and rolled up to the customer level using their single best visit — while hard business facts like “added to cart” always take priority over the behavioral score for any operational decision.

2. The Data Layer — What We Capture

A lightweight tracking script on the storefront records visitor behavior and sends it to a dedicated events database. A background ingestion process reads that stream continuously and writes it into three clean, structured tables inside each brand's own database:

Table	What it holds
behavioral_events	Every meaningful action: page views, product views, add-to-cart, checkout started, and scroll depth — with the page URL, product, and timing attached.
click_events	Every click, already classified as a useful click or a dead click (a click that led nowhere), with the element clicked and where on the page.
intent_sessions	One row per browsing session: how long it lasted, how many of each event type it had, and the order those events happened in.

This ingestion runs every 15 minutes, only ever processing what's new or changed since the last run (never re-reading the entire history), and is designed to be safe to re-run — seeing the same event twice never creates a duplicate or double-counts anything.

3. Analytics Rollups — Daily Engagement Summaries

On top of the raw tables, five daily summary tables are maintained automatically, so common business questions can be answered instantly without scanning raw data. Each one recomputes its own numbers fresh whenever new or corrected data arrives for a given day, so the figures are always self-consistent.

Rollup	What it tells you
Daily Summary	The single top-line view: sessions, unique visitors, page/product views, clicks, and add-to-cart/checkout counts for the whole brand, per day.
Page Behavior	Which pages get engagement, which generate dead clicks, and which pages are associated with add-to-cart or checkout activity.
Click Behavior	Which specific UI elements — buttons, links, page sections — produce useful interactions versus dead clicks, so UX friction is easy to spot.
Product Behavior	Per-product views, add-to-carts, and conversion-style rates, so product performance can be compared day over day.
Behavioral Paths	Common journeys through the site (e.g. viewed a product, clicked, added to cart) and how often each journey leads to a real add-to-cart or checkout.

Why this matters before scoring

These rollups are what make the intent scoring engine possible: by the time scoring runs, the system already knows what a “normal” day of traffic looks like for that brand, which is exactly what the scoring engine needs to judge any individual session against.

4. The Intent Scoring Engine

This is the predictive layer. It looks at a finished (closed) browsing session and answers two separate questions: **how engaged was this visit, behaviorally?** and, independently, **does a hard business fact mean we should treat this as high-intent regardless?**

4.1 Two Jobs, on Purpose

The engine runs as two separate, independently scheduled jobs, running in their own dedicated container so they never compete with ingestion for resources.

Job	Runs	Purpose
Intent Scoring	Every 2 hours	Scores newly-closed sessions using the thresholds already known for that day. Never changes those thresholds.
Daily Calibration	Once daily, 02:00	Once a day is fully over, works out what “Low / Medium / High” actually means for that specific day's traffic, and fixes it in place permanently.

This split exists so that a session scored early in the day isn't silently reclassified later just because more traffic showed up afterward — the classification for a given day only settles once, the day after, using that day's complete picture.

4.2 Who Gets Excluded

- **Bots and crawlers** — identified from known crawler signatures (search engine bots, scraping tools, headless browsers). Excluded entirely, with no score at all.

- **True zero-signal visits** — a session is only excluded as “nothing happened” when it had one event or fewer, under a second of recorded time, zero clicks, *and* zero scrolling, all at once. A short or 0-second duration alone is never enough by itself, since some legitimate sessions sync in batches and can show 0ms while still containing real activity.
- **Sessions still in progress** — a session with no recorded duration yet simply isn't scored until it actually closes. It's picked up automatically the moment it does.

4.3 How the Score Is Built

Five behavioral signals are measured, each capped so no single one can dominate the result, then combined into a single weighted score:

Signal	Capped at	Weight
Products viewed	4	11.0 pts each
Useful clicks	12	2.2 pts each
Minutes spent	6	4.5 pts each
Scroll events	8	1.8 pts each
Pages viewed	6	1.5 pts each

Two adjustments are then applied: a **struggle penalty** if the visitor had a high dead-click rate compared to almost everyone else that day, and a **bounce penalty** for very short or near-empty sessions. The result is then multiplied by a page-type factor — reaching a checkout or product page counts for more than just landing on the home page — and finally capped to a clean 0–100 scale. That final number is the **predictive score**.

4.4 Turning a Score into Low / Medium / High

The score is compared against that specific day's own distribution of scores — not a fixed number that never changes. The bottom third or so of that day's sessions become **Low**, the top slice becomes **High**, and everything in between is **Medium**. On days with too little traffic to calculate this reliably, a safe fallback split is used instead, and the system records which method was used.

4.5 Hard Overrides — Facts Beat Behavior

A session's behavioral score and its final operational classification are kept as two separate fields, on purpose. Regardless of how the behavioral score comes out, the session is always treated as operationally **High** if any of the following happened:

- The visit landed directly on the checkout page (for example, someone clicking a checkout-recovery link).
- A checkout was started.
- Something was added to the cart.

This means a session can have a low behavioral score and still be flagged operationally High — the engine keeps both facts, rather than letting one silently overwrite the other.

5. Rolling It Up to the Customer

Every visitor is also given a single, customer-level classification, using one simple rule:

An actor's intent is their single best session

If a customer's visits scored Low, Low, and High, that customer's overall classification is **High**. One standout, highly engaged visit is treated as a stronger signal than several average ones — it is deliberately not an average, and not just their most recent visit.

If every visit a customer has ever made turns out to be excluded (bot traffic, or genuinely empty visits), that customer is left unclassified rather than being defaulted to Low — “we have no real signal on this person” is treated as its own honest outcome, not guessed at.

Customer-level records are refreshed both when one of their sessions is newly scored, and whenever the daily calibration step changes how an existing session is classified — so a customer's record always reflects their true best visit, even if that visit happened earlier in the day before the day's numbers were finalized.

Field	What it captures
Predictive score	Score of this customer's single best session.
Intent bucket	Low / Medium / High, from that best session.
Best session	Which specific visit earned this classification.
Session count	How many of this customer's visits have been scored so far.

6. How It's Deployed

The system runs as two independent, always-on services sharing the same brand databases, so a problem in one never blocks the other.

Service	Responsible for	Schedule
Ingestion & Rollups	Pulling raw events and maintaining the five daily analytics tables	Every 15 minutes
Intent Scoring Worker	Scoring closed sessions and daily threshold calibration	Every 2 hours / daily at 02:00

Both services also run their jobs once immediately whenever they start or restart, so a redeploy never leaves data stale until the next scheduled tick. Every write anywhere in the system is designed to be safely repeatable — running the same job twice, or over overlapping time windows, always produces the same result, never a duplicate or an over-count.

7. What's Provisional in This Version

This is a first, deliberately scoped version. A few things are known, documented starting points rather than fully tuned business rules:

- The page-type scoring factors and the fallback Low/Medium/High cut-offs are reasonable starting values, not numbers derived from this brand's historical conversion data yet.

- The bot-detection list covers well-known crawlers and automation tools, and can be extended at any time as new ones are identified.
- If a day's calibration step is ever missed, that day quietly falls back to the most recent earlier calibration rather than being automatically retried.
- The system does not yet distinguish a useful click from a dead click *within* a customer's journey path — only that a click happened.

Bottom line

The foundation — clean data capture, reliable daily analytics, and a fully automated, self-correcting scoring pipeline — is in place and running continuously. The specific numeric weights are the intended next thing to calibrate against real outcomes.